

- Description

This module manages a smart greenhouse CO₂ monitoring and ventilation control system, ensuring that:

1. The system continuously measures the CO₂ concentration in the greenhouse (in ppm).
2. If the CO₂ level exceeds a predefined maximum threshold, the ventilation system is activated to reduce CO₂ concentration.
3. If the CO₂ level falls below a predefined minimum threshold, the ventilation system is reduced or turned off to conserve CO₂ for optimal plant growth.
4. If the CO₂ level is within the optimal range, the ventilation system remains at a default setting.

This system helps maintain optimal CO₂ levels for plant growth, ensures energy efficiency, and optimizes greenhouse conditions.

- Specification

module SmartGreenhouseCO2Control

var currentCO2, minCO2, maxCO2: real

var currentVentSetting, defaultVentSetting: real

process updateCO2(newCO2: real) returns (currentCO2: real, currentVentSetting: real)

functional scenario form:

currentCO2 > maxCO2 && currentVentSetting = 100.0, output "High CO₂: " + currentCO2 + " ppm. Ventilation set to 100%.", Ventilation set to maximum (100%).

||

currentCO2 < minCO2 && currentVentSetting = 0.0, output "Low CO₂: " + currentCO2 + " ppm. Ventilation set to 0%.", Ventilation set to minimum (0%).

||

currentCO2 >= minCO2 && currentVentSetting = defaultVentSetting, output "CO₂ optimal: " + currentCO2 + " ppm. Ventilation remains at " + defaultVentSetting + "%.", Ventilation remains at default setting.

end process

end module

- Java Code

```
package code;
public class c90_SmartGreenhouseCO2 {
    private double minCO2;
    private double maxCO2;
    private double currentCO2;
    private double defaultVentSetting;
    private double currentVentSetting;
    public c90_SmartGreenhouseCO2(double minCO2, double maxCO2, double
initialCO2, double defaultVentSetting) {
```

```

        this.minCO2 = minCO2;
        this.maxCO2 = maxCO2;
        this.currentCO2 = initialCO2;
        this.defaultVentSetting = defaultVentSetting;
        this.currentVentSetting = defaultVentSetting;
    }

    public void updateCO2(double newCO2) {
        currentCO2 = newCO2;
        adjustVentilation();
        System.out.printf("Current CO2 Level: %.2f ppm, Ventilation Setting: %.2f%%\n", currentCO2, currentVentSetting);
        if (currentCO2 > maxCO2) {
            System.out.println("ALERT: High CO2 concentration detected! Increasing ventilation.");
        } else if (currentCO2 < minCO2) {
            System.out.println("ALERT: Low CO2 concentration detected! Reducing ventilation.");
        } else {
            System.out.println("CO2 level is optimal. Ventilation remains at default setting.");
        }
    }

    private void adjustVentilation() {
        if (currentCO2 > maxCO2) {
            currentVentSetting = 100.0;
        } else if (currentCO2 < minCO2) {
            currentVentSetting = 0.0;
        } else {
            currentVentSetting = defaultVentSetting;
        }
    }

    public double getCurrentCO2() {
        return currentCO2;
    }

    public double getCurrentVentSetting() {
        return currentVentSetting;
    }

    public static void main(String[] args) {
        c90_SmartGreenhouseCO2 greenhouse = new c90_SmartGreenhouseCO2(300.0, 800.0, 500.0, 50.0);
        // testcase-VT:
        //     greenhouse.updateCO2(850.0);
        //     greenhouse.updateCO2(250.0);
        //     greenhouse.updateCO2(600.0);
    }

```

```

        // testcase-FT:
//      greenhouse.updateCO2(-1.7976931348623157E308);
//      greenhouse.updateCO2(1.7976931348623157E308);
//      greenhouse.updateCO2(0.0);

        // testcase-Z3:
//      greenhouse.updateCO2(850.0);
//      greenhouse.updateCO2(250.0);
//      greenhouse.updateCO2(600.0);
        // testcase-UVT:
        greenhouse.updateCO2(850.0);
        greenhouse.updateCO2(250.0);
        greenhouse.updateCO2(600.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c90_SmartGreenhouseCO2.java	100.0 %	8	0	8
▼ c90_SmartGreenhouseCO2	100.0 %	8	0	8
main(String[])		0	0	0
c90_SmartGreenhouseCO2(doul		0	0	0
adjustVentilation()	100.0 %	4	0	4
updateCO2(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	108	21,506	21,614
▼ src	0.5 %	108	21,506	21,614
▼ code	0.5 %	108	21,506	21,614
▼ c90_SmartGreenhouseCO2.java	94.7 %	108	6	114
▼ c90_SmartGreenhouseCO2	94.7 %	108	6	114
main(String[])	100.0 %	18	0	18
c90_SmartGreenhouseCO2(doul	100.0 %	18	0	18
adjustVentilation()	100.0 %	25	0	25
updateCO2(double)	100.0 %	47	0	47

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c90_SmartGreenhouseCO2.java	75.0 %	6	2	8
▼ c90_SmartGreenhouseCO2	75.0 %	6	2	8
main(String[])		0	0	0
c90_SmartGreenhouseCO2(doul		0	0	0
adjustVentilation()	75.0 %	3	1	4
updateCO2(double)	75.0 %	3	1	4

5. Fuzz testing instruction coverage: 88.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	101	21,513	21,614
▼ src	0.5 %	101	21,513	21,614
▼ code	0.5 %	101	21,513	21,614
▼ c90_SmartGreenhouseCO2.java	88.6 %	101	13	114
▼ c90_SmartGreenhouseCO2	88.6 %	101	13	114
main(String[])	100.0 %	18	0	18
c90_SmartGreenhouseCO2(double, c90_SmartGreenhouseCO2)	100.0 %	18	0	18
adjustVentilation()	84.0 %	21	4	25
updateCO2(double)	93.6 %	44	3	47

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c90_SmartGreenhouseCO2.java	100.0 %	8	0	8
▼ c90_SmartGreenhouseCO2	100.0 %	8	0	8
main(String[])		0	0	0
c90_SmartGreenhouseCO2(double, c90_SmartGreenhouseCO2)		0	0	0
adjustVentilation()	100.0 %	4	0	4
updateCO2(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	108	22,921	23,029
▼ src	0.5 %	108	22,921	23,029
▼ code	0.5 %	108	22,921	23,029
▼ c90_SmartGreenhouseCO2.java	94.7 %	108	6	114
▼ c90_SmartGreenhouseCO2	94.7 %	108	6	114
main(String[])	100.0 %	18	0	18
c90_SmartGreenhouseCO2(double, c90_SmartGreenhouseCO2)	100.0 %	18	0	18
adjustVentilation()	100.0 %	25	0	25
updateCO2(double)	100.0 %	47	0	47

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c90_SmartGreenhouseCO2.java	100.0 %	8	0	8
▼ c90_SmartGreenhouseCO2	100.0 %	8	0	8
main(String[])		0	0	0
c90_SmartGreenhouseCO2(double, c90_SmartGreenhouseCO2)		0	0	0
adjustVentilation()	100.0 %	4	0	4
updateCO2(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	108	22,921	23,029
▼ src	0.5 %	108	22,921	23,029
▼ code	0.5 %	108	22,921	23,029
▼ c90_SmartGreenhouseCO2.java	94.7 %	108	6	114
▼ c90_SmartGreenhouseCO2	94.7 %	108	6	114
main(String[])	100.0 %	18	0	18
c90_SmartGreenhouseCO2(double, c90_SmartGreenhouseCO2)	100.0 %	18	0	18
adjustVentilation()	100.0 %	25	0	25
updateCO2(double)	100.0 %	47	0	47